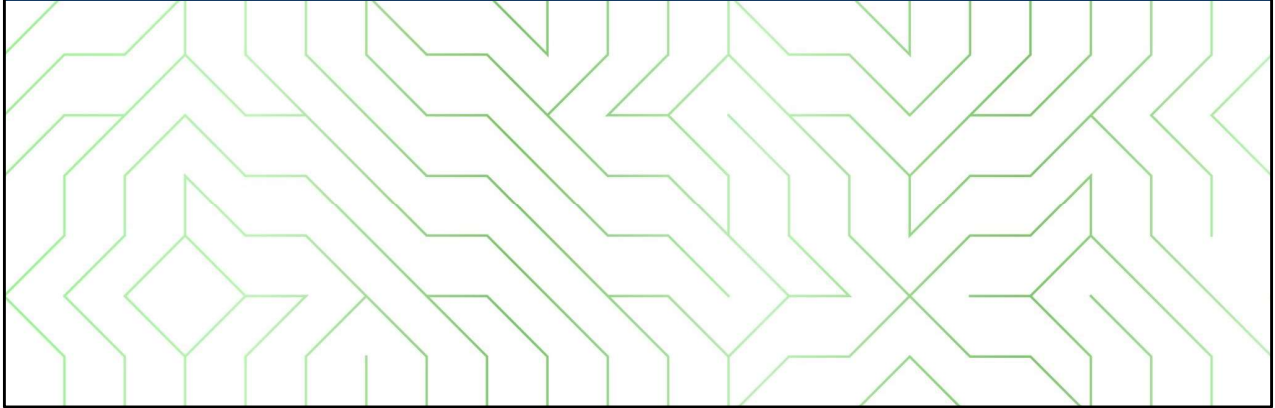

Cerebrovascular Disease

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IS-MSN Fall 2021



Learning Objectives

- **Acute Ischemic Stroke:**
 - Appreciate and be able to counsel your patients regarding risk of stroke based on risk factors
 - Participate in the evaluation and treatment of patients presenting with acute stroke syndromes.
 - Initiate therapies for secondary stroke prevention.
- **Transient Ischemic Attack**
 - Understand the work up for a patient presenting with transient neurologic deficits and differentiate from acute stroke syndrome.
 - Initiate therapies to reduce risk of future stroke.
- **Cardioembolic stroke**
 - Appreciate the tell-tale signs of a cardioembolic stroke based on risk factors, history and diagnostic work up.
 - Understand benefits and potential risks for treatment depending on cause and stratification of this patient population.
- **Intracerebral Hemorrhage**
 - Know the most common causes and their differentiating characteristics.

Learning Objectives

- **Subarachnoid Hemorrhage:**
 - Recognize clinical presentation of subarachnoid hemorrhage
 - Understand the different types of subarachnoid hemorrhage and risk factors
 - Initiate therapies for hemorrhage and prevention of further bleeding
- **Venous Sinus Thrombosis**
 - Understand the diverse clinical presentation of DVST
 - Recognize significant risk factors, most susceptible populations.
 - Initiate therapies to reduce morbidity and mortality.

Definitions

What is a Stroke?

→ **Lack of oxygen** to the brain or spinal cord resulting in ischemia and/or infarction

→ **Causes:**

→ **Ischemic stroke**

→ Can be 'silent' or without symptoms

→ **Hemorrhagic stroke**

→ Intracerebral hemorrhage

→ Subarachnoid hemorrhage

→ **Cerebral venous thrombosis**

→ Other: "Stroke, NOS"

Table 1. Definition of Stroke

The term "stroke" should be broadly used to include all of the following: Definition of CNS infarction: CNS infarction is brain, spinal cord, or retinal cell death attributable to ischemia, based on: 1. pathological, imaging, or other objective evidence of cerebral, spinal cord, or retinal focal ischemic injury in a defined vascular distribution; or 2. clinical evidence of cerebral, spinal cord, or retinal focal ischemic injury based on symptoms persisting ≥24 hours or until death, and other etiologies excluded. (Note: CNS infarction includes hemorrhagic infarctions, types I and II; see "Hemorrhagic infarction.") Definition of ischemic stroke: An episode of neurological dysfunction caused by focal cerebral, spinal, or retinal infarction. (Note: Evidence of CNS infarction is defined above.) Definition of silent CNS infarction: Imaging or neuropathological evidence of CNS infarction, without a history of acute neurological dysfunction attributable to the lesion. Definition of intracerebral hemorrhage: A focal collection of blood within the brain parenchyma or ventricular system that is not caused by trauma. (Note: Intracerebral hemorrhage includes parenchymal hemorrhages after CNS infarction, types I and II—see "Hemorrhagic infarction.") Definition of stroke caused by intracerebral hemorrhage: Rapidly developing clinical signs of neurological dysfunction attributable to a focal collection of blood within the brain parenchyma or ventricular system that is not caused by trauma. Definition of silent cerebral hemorrhage: A focal collection of chronic blood products within the brain parenchyma, subarachnoid space, or ventricular system on neuroimaging or neuropathological examination that is not caused by trauma and without a history of acute neurological dysfunction attributable to the lesion. Definition of subarachnoid hemorrhage: Bleeding into the subarachnoid space (the space between the arachnoid membrane and the pia mater of the brain or spinal cord). Definition of stroke caused by subarachnoid hemorrhage: Rapidly developing signs of neurological dysfunction and/or headache because of bleeding into the subarachnoid space (the space between the arachnoid membrane and the pia mater of the brain or spinal cord), which is not caused by trauma. Definition of stroke caused by cerebral venous thrombosis: Infarction or hemorrhage in the brain, spinal cord, or retina because of thrombosis of a cerebral venous structure. Symptoms or signs caused by reversible edema without infarction or hemorrhage do not qualify as stroke. Definition of stroke, not otherwise specified: An episode of acute neurological dysfunction presumed to be caused by ischemia or hemorrhage, persisting ≥24 hours or until death, but without sufficient evidence to be classified as one of the above. CNS indicates central nervous system.	
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Time is brain...

- A **sudden onset of neurological deficit** is the unifying concept in stroke
- Stroke is a clinical diagnosis
 - Rapid bedside assessment establishes a diagnosis of stroke but not underlying mechanism
 - To distinguish among different mechanisms requires imaging and sometimes laboratory workup
- Ischemia refers to inadequate blood flow, causes include:
 - Embolism
 - Thrombosis
 - Hemorrhage
- **Infarct is cell death, time is brain**

*Different strokes
for different folks*

Ischemic stroke

- Large Vessel Occlusion
 - Cardioembolic
 - Artery-Artery embolism
- Small Vessel Occlusion

Hemorrhagic stroke

- Intracerebral hemorrhage
- Subarachnoid hemorrhage

Venous thrombosis

Epidemiology

Condition	2000	2015	2016
All Causes	869	733	728.8
Heart Dz	257.6	168.5	165.5
Cancer	199.6	158.5	155.8
Chronic Lower Resp	44.2	41.6	40.6
Unintent. Injuries	34.9	43.2	47.4
STROKE	60.9 (RANK 3)	37.6	37.3 (RANK 5)
Alzheimer Dz	18.1	29.4	33.3
Diabetes	25	21.3	21
Influenza/Pneum.	23.7	15.2	13.5
Nephritis	13.5	13.4	13.1
Suicide	10.4	13.3	13.5

- Stroke is the second most common cause of mortality and third most common cause of disability worldwide.
- Globally, 68% of all strokes are ischemic and 32% are hemorrhagic.
- In the US:
 - 5TH Leading Cause of Death in US (2016)
 - 87% of all strokes are ischemic
 - 10% hemorrhagic
 - 3% being subarachnoid hemorrhage
 - A stroke occurs every 40 seconds in the US
 - Every 3 minutes and 42 seconds a death from stroke occurs in US (2016)

Acute Ischemic Stroke and TIA

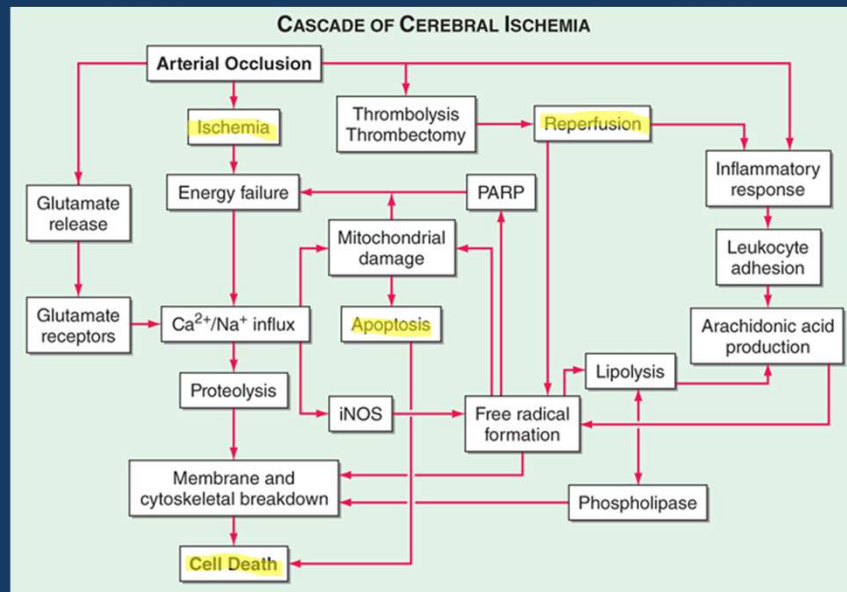
Ischemic Stroke

Large Vessel Occlusion

- Due to thrombotic or embolic occlusion of the major arteries of the brain like the internal carotid artery, middle cerebral artery, anterior cerebral artery or the vertebrobasilar system
- Cardioembolic stroke
 - Blood clot, air, septic debris, or fat is launched into the cerebral vasculature by the heart

Small Vessel or Lacunar Infarct

- Local (in situ) thrombosis within the small arteries causes infarct directly in the tissue where it starts



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition
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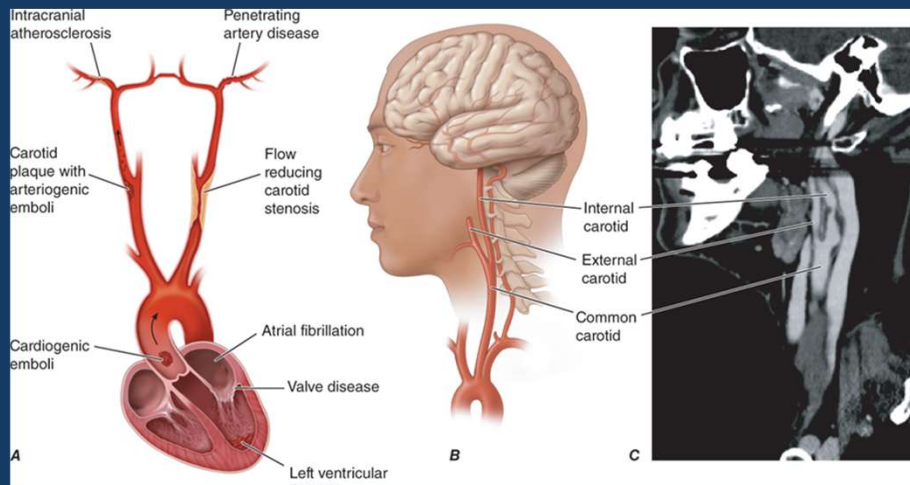
Cascade of Cerebral Ischemia.

iNOS- inducible nitrous oxide synthetase

PARP- poly-A-ribose polymerase

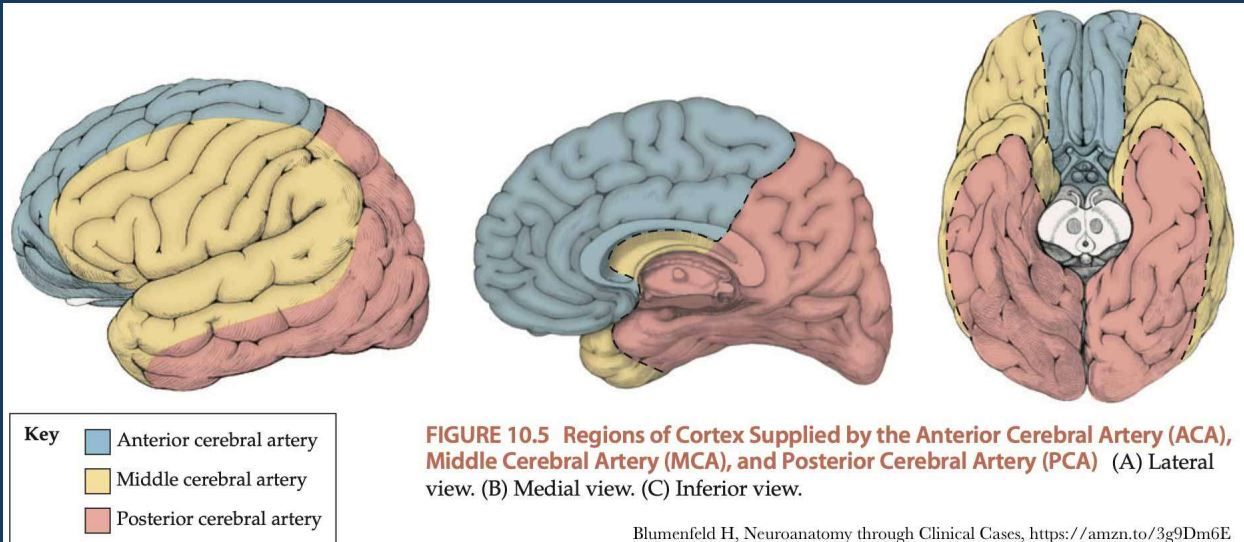
Neurons are starved of glucose and O₂, which results in mitochondrial failure to produce ATP. Without ATP, membrane ion pumps stop functioning and neurons depolarize, allowing intracellular CA to rise. Cellular depolarization causes glutamate release from synaptic terminals, then activating post-synaptic glutamate receptors, which increase CA influx, which is neurotoxic. Free radicals produce degradation of membrane lipids and mitochondrial dysfunction; cause destruction of membranes.

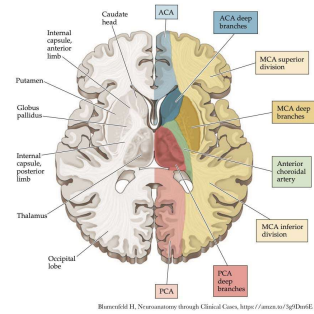
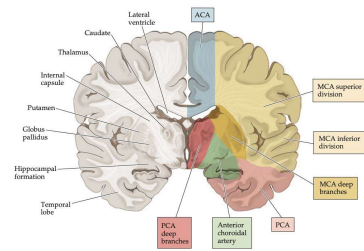
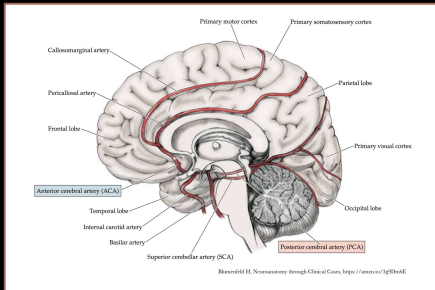
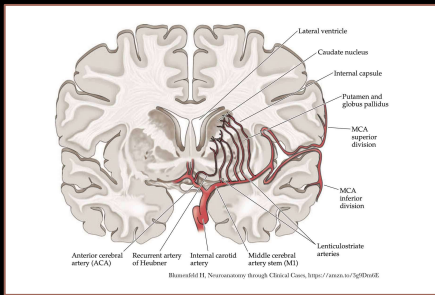
Pathophysiology



Pathophysiology of ischemic stroke. A. Diagram illustrating the three major mechanisms that underlie ischemic stroke: (1) occlusion of an intracranial vessel by an embolus that arises at a distant site (e.g., cardiogenic sources such as atrial fibrillation or artery-to-artery emboli from carotid atherosclerotic plaque), often affecting the large intracranial vessels; (2) in situ thrombosis of an intracranial vessel, typically affecting the small penetrating arteries that arise from the major intracranial arteries; (3) hypoperfusion caused by flow-limiting stenosis of a major extracranial (e.g., internal carotid) or intracranial vessel, often producing “watershed” ischemia. B. and C. Diagram and reformatted computed tomography angiogram of the common, internal, and external carotid arteries. High-grade stenosis of the internal carotid artery, which may be associated with either cerebral emboli or flow-limiting ischemia, was identified in this patient.

Vascular territories





Mechanisms of acute ischemic stroke

- Vascular thrombosis
 - Atherosclerosis (extracranial or intracranial), dissection (spontaneous or traumatic), vasculopathies (Moyamoya, fibromuscular dysplasia, vasculitis)
- Embolic etiologies
 - Atherosclerosis (**proximal artery embolism**), **cardioembolic** (afib, endocarditis, PFO)
- Hypercoagulable states affecting arteries
 - Malignancy, antiphospholipid syndrome
- Venous hypertension, hypercoagulable states affecting veins
 - Dural sinus thrombosis due to malignancy, antiphospholipid syndrome, factor V leiden, protein C or S deficiency
 - Paradoxical embolism: DVT or another venous thrombus makes its way through a PFO to the left side of the heart to cause a cardioembolic stroke

Risk Factors

→ Non-Modifiable Risk Factors Include:

- Age
- Race
- Sex
- Ethnicity
- Family history of stroke or transient ischemic attacks (TIAs)

→ Modifiable Risk Factors Include:

- Hypertension
- Diabetes mellitus
- Cardiac disease
- High cholesterol
- **Carotid stenosis**
- **Atrial Fibrillation**
- Lifestyle issues: Excessive alcohol intake, tobacco use, illicit drug use, physical inactivity
- Obesity
- Oral contraceptive use/postmenopausal hormone use (probably estrogen more than progesterone)

Clinical Presentation

- Many different clinical presentations, depending on location of the stroke
- Examples of chief complaints:
 - Weakness
 - Numbness
 - Difficulty speaking
 - Difficulty walking
 - Confusion
 - 'not right'
- Stroke should be considered in any **sudden onset neurologic deficit**
- **Your stroke syndromes lecture focused on combinations of deficits commonly seen in stroke, today we're more interested in how to proceed after the diagnosis is made**

Highly variable dependent on location of the infarct. AIS and hemorrhagic stroke can be clinically indistinguishable. Your localization skills are important here. Symptoms and exam should make physiologic sense, but don't be too rigid with this as you can miss rare stroke syndromes.

Differential Diagnosis Considerations

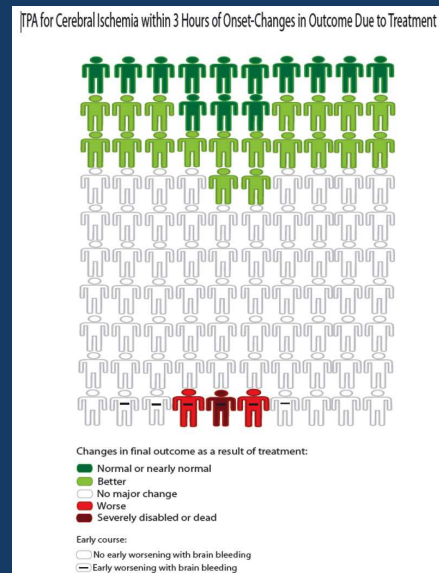
- Consider different mechanisms of stroke
- Transient Ischemic Attack (TIA)
- Seizure and post-ictal paralysis
- Migraine with Aura
- Functional neurologic deficit
- Cerebral Amyloid Angiopathy

Treatment/management principles

- Time is Brain
 - Once acute ischemic stroke is suspected, rapid determination of eligibility for acute interventions is needed
 - Every minute brain is deprived of blood flow leads to irreversible ischemia and neuronal death.
- Acute interventions for ischemic stroke include:
 - Acute thrombolysis with **IV tPA**
 - Mechanical **Thrombectomy** for large vessel occlusion
 - Patients may get both, one or neither of these interventions depending on many factors

IV tPA – Mechanism of action and establishment of efficacy

- Binds to fibrin within a clot
- Converts entrapped plasminogen to plasmin
- Proven to improve outcomes when given in accordance with strict inclusion/exclusion criteria
 - NINDS 1995
 - ECASS 2008
 - SITS-MOST 2007



Tissue plasminogen activator for acute ischemic stroke. The National Institute of Neurological Disorders and Stroke rt-PA Stroke Study Group. *N Engl J Med* 1995;333(24):1581–1587. doi:10.1056/NEJM199512143332401.

Hacke W, Kaste M, Bluhmki E, et al. Thrombolysis with alteplase 3 to 4.5 hours after acute ischemic stroke. *N Engl J Med* 2008;359(13):1317–1329. doi:10.1056/NEJMoa0804656.

Wahlgren N, Ahmed N, Dávalos A, et al. Thrombolysis with alteplase for acute ischaemic stroke in the Safe Implementation of Thrombolysis in Stroke-Monitoring Study (SITS-MOST): an observational study. *Lancet* 2007;369(9558): 275–282. doi:10.1016/S0140-6736(07)60149-4.

Acute thrombolysis with IV tPA: eligibility algorithm

- Establish a **last known well time (LKWT)**
- Focused neurological exam with **National Institutes of Health Stroke Scale (NIHSS)**
- Brief medication history (e.g., anticoagulants) to assess contraindications to therapy
- **Non contrast head CT** to rule out hemorrhage
- Labs: **fingerstick blood glucose**, coagulation studies such as PT-INR, PTT, platelet count, and renal functions such as creatinine
 - Fingerstick glucose: technically only lab REQUIRED before IV tPA can be administered (caveats apply)
 - Coagulation studies: abnormalities on these labs can be exclusion criteria for IV tPA

Physical Examination

- Urgency of the evaluation will dictate the level of detail in your exam
- NIHSS – national institutes of health stroke scale
 - Standardized neurologic assessment for high yield stroke exam findings
 - All providers should have training on NIHSS
 - Scored 0 to 42 (higher is worse)
 - Reliable and validated screening tool for stroke assessment
 - Less sensitive to posterior circulation stroke syndromes

If you will be doing a neurology, ER, or critical care rotation, please [visit this website](#) and complete your NIHSS certification. It will be invaluable when you get to third year.

CATEGORY	STROKE SCALE	
1a. Level of consciousness Alert, Drowsy, etc	0 Alert 1 Drowsy 2 Stuporous 3 Coma	6.a Motor Leg - Left Elevate extremity to 30 degrees and score drift/movement. Count to 5 out loud and use fingers for visual cue. 0 No Drift 1 Drift 2 Can't resist gravity 3 No effort against gravity 4 No Movement NT = Amputation, joint fusion
1b. LOC Questions Month, age	0 Answers both correctly 1 Answers one correctly 2 Incorrect	6.b Motor Leg - Right Elevate extremity to 30 degrees and score drift/movement. Count to 5 out loud and use fingers for visual cue. 0 No Drift 1 Drift 2 Can't resist gravity 3 No effort against gravity 4 No Movement NT = Amputation, joint fusion
1c. LOC Commands Open/close eyes, make a fist & let go	0 Obays both correctly 1 Obays one correctly 2 Incorrect	7. Limb Ataxia Finger to nose, heel down shin 0 Absent 1 Present in one limb 2 Present in two limbs
2. Best Gaze Eyes open - pt follows examiner's fingers or face.	0 Normal 1 Partial gaze palsy 2 Partial deviation	8. Sensory Pin prick to face, arms, trunk, and legs - compare sharpness side to side 0 Normal 1 Partial loss 2 Severe Loss
3. Visual Introduce visual stimulus/threat to pt's visual field quadrants. Cover 1 eye and hold up fingers in all 4 quadrants.	0 No visual loss 1 Partial hemianopia 2 Complete hemianopia 3 Bilateral hemianopia	9. Best Language Name items, describe picture, and read sentences. Don't forget glasses if they normally wear them. 0 No aphasia 1 Mild to moderate aphasia 2 Severe aphasia 3 Mute
4. Facial Palsy Show teeth, raise eyebrows and squeeze eyes tightly shut.	0 Normal 1 Minor 2 Partial 3 Complete	10. Dysarthria Evaluate speech clarity by pt reading or repeating words on list 0 Normal articulation 1 Mild to moderate dysarthria 2 Near to unintelligible or worse NT = Intubated or other physical barrier
5.a Motor Arm - Left Elevate extremity to 90 degrees and score drift/movement. Count to 10 out loud and use fingers for visual cue.	0 No Drift 1 Drift 2 Can't resist gravity 3 No effort against gravity 4 No Movement NT = Amputation, joint fusion	11. Extinction and Inattention Use information from prior testing or double simultaneous stimuli testing to identify neglect. Face, arms, legs, and visual fields. 0 No neglect 1 Partial neglect 2 Complete neglect
5.b Motor Arm - Right Elevate extremity to 90 degrees and score drift/movement. Count to 10 out loud and use fingers for visual cue.	0 No Drift 1 Drift 2 Can't resist gravity 3 No effort against gravity 4 No Movement NT = Amputation, joint fusion	

NT = Not Testable acceptable as noted above

NIHSS Instruction

1. Administer stroke scale items in the order listed.
2. Record performance in each category after each subscale exam.
3. Do not go back and change scores.
4. Follow directions provided for each exam technique.
5. Scores should reflect what the patient does, not what the clinician thinks the patient can do.
6. The clinician should record answers while administering the exam and work quickly.
7. Except where indicated, the patient should not be coached (i.e., repeated requests to patient to make a special effort).

Initial Imaging in Acute Stroke

Always:

- CT Head wo contrast
 - Weakness for posterior fossa strokes

Sometimes:

- CT Angiogram
- CT Perfusion
- MRI Brain wo contrast
 - MR Angiogram
 - MR Perfusion

STROKE



*probably not a tPA candidate

Imaging in acute stroke

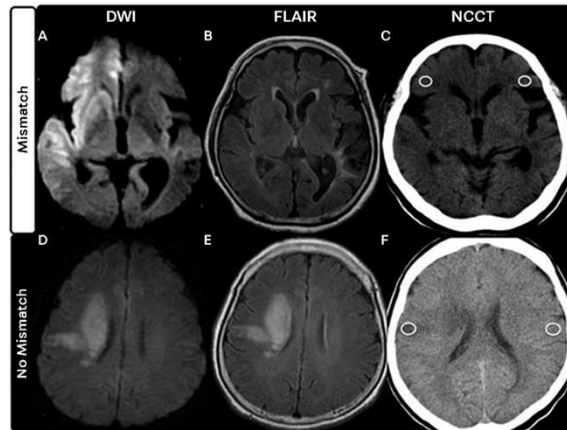


FIGURE 3-7

Diffusion-weighted imaging (DWI) and fluid-attenuated inversion recovery (FLAIR) MRI changes with corresponding changes on noncontrast CT (NCCT). When a mismatch is seen between DWI and FLAIR, noncontrast CT shows only subtle early ischemic changes (A, B, and C). When no mismatch is seen between DWI and FLAIR, noncontrast CT shows subacute-appearing changes (D, E, and F).

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KEY POINT

● On noncontrast CT, brain regions that are darker than the contralateral normal-appearing white matter are a marker of increased risk of hemorrhage with thrombolysis. The presence of these regions does not mean that thrombolysis is absolutely contraindicated, but they do mean that clinicians should proceed with caution after weighing the risks and benefits of thrombolysis.

History

Step 1: Establish the time of onset of stroke which starts the clock on all subsequent time windows and management decisions.

- “When was the patient last seen well?”
- “When was the patient last seen normal?”
- “When was the patient last known well?” (**LKW**)
- Write down the time, ex: 8:45 am; not “45 minutes ago”.

Step 2: History of the presenting symptom (which will help distinguish mimics from stroke). Temporal onset needed. Stroke is “abrupt”, not gradual like migraine for instance. Are symptoms “stuttering”, fluctuating, improving?

Step 3: PMHX (to determine risk factors and eligibility for IV rtPA)

Hx of anticoagulants, recent trauma, bleeding, recent hospitalizations. May need family member to answer the questions.

Treatment algorithm

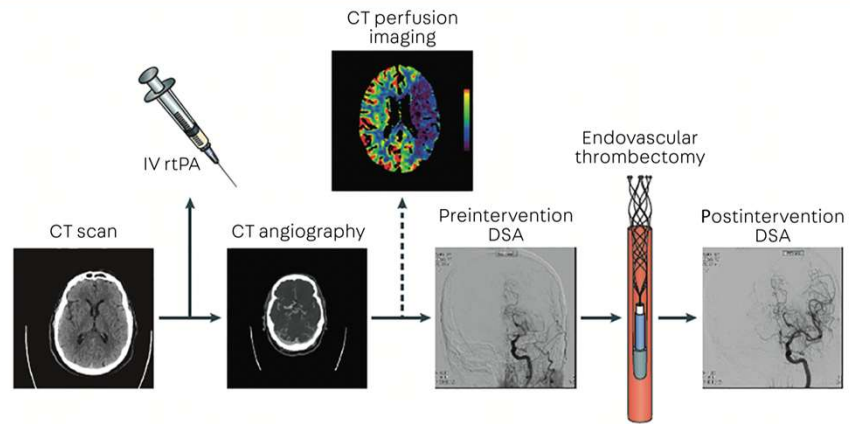


FIGURE 2-3

Graphic illustrating the sequence of steps in contemporary acute stroke therapy. IV thrombolysis should optimally be given to all patients without contraindications within 60 minutes of their arrival to the emergency department. Ideally, the time from the qualifying scan to the groin puncture in candidates for endovascular therapy should be shorter than 60 minutes. The use of perfusion imaging is optional within the first 6 hours after symptom onset, but it becomes necessary to select patients for endovascular therapy between 6 and 24 hours after last known well.

CT = computed tomography; DSA = digital subtraction angiography; IV = intravenous; rtPA = recombinant tissue plasminogen activator.

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Acute thrombolysis with IV tPA

- Window to administer IV tPA: **within 4.5 hours from LKWT**
- Benefit of tPA when given within 3 hours of LKWT:
 - 13% return to baseline
 - 19% will have some functional improvement
- Risks of tPA when given within 3 hour window:
 - 6% chance of hemorrhage
- Data supports use of tPA in extended window of **3-4.5 hours from LKWT in select patients:**
 - Age >18 yo and <80 yo
 - NIHSS <25
 - Ischemic injury on CT <1/3 of MCA territory
 - Excludes patients with history of stroke AND diabetes

Contraindications to IV tPA

- Absolute contraindications
 - CT shows hemorrhage or large hypodensity in the area with suspected
 - Treatment with anticoagulation
 - INR >1.7 or platelet count <100,000
 - Blood glucose <50 mg/dL
 - Significant head trauma, intracranial or intraspinal surgery within previous 3 months
 - Recent LP
 - Blood pressure sustained over 185/110 mmHg
- Relative contraindications
 - Pregnancy
 - Stroke or myocardial infarction in previous 3 months
 - Major trauma in previous 14 days
 - History of ICH, unruptured and unsecured intracranial aneurysm, AVM, or intracranial neoplasm
 - Arterial puncture in previous 7 days

Post IV tPA management

- Admission to ICU for close neurological monitoring
 - Neurological and BP checks every 15 minutes during infusion, every 30 min for 6 hours after, and every hour for 24 hours
- No antiplatelet agents, anticoagulation, or DVT prophylaxis for 24 hours after tPA
- Blood pressure goal **SBP <180, DBP <105**
- Complications:
 - Any acute neurological change, severe headache or vomiting should prompt repeat CT brain to assess for hemorrhagic transformation

*Mechanical
Thrombectomy for
Large Vessel
Occlusion*



Mechanical Thrombectomy for Large Vessel Occlusion

- Patients should be considered for mechanical thrombectomy **within 6 hours of LKWT** if there is a **large vessel occlusion in the anterior circulation** that is accessible by catheterization (carotid terminus, M1 or proximal M2 segment of MCA)
- Perfusion imaging (CT or MRI) provides information on infarct and potentially salvageable tissue, characterized by mismatch ratio and volume of infarct
- Patients who have completed their infarcts (no salvageable tissue on imaging) are unlikely to benefit from thrombectomy
- Several studies have shown benefit of thrombectomy in addition to tPA over IV tPA alone – i.e. **you can get thrombectomy even if you already got tPA if you meet criteria**
- **NNT of 2.6!!!!**

Extended Window for Thrombectomy

- DAWN trial and DEFUSE3 trials used perfusion imaging to identify patients >6 hours, and less than 24 hours (DAWN) or 16 hours (DEFUSE3) that might benefit from thrombectomy at even greater time periods
- These studies showed benefits, with NNT between 2 – 4.
- While not completely standard of care at this point, most large comprehensive stroke centers will perform thrombectomy up to 24 hours in the right situation
- Important for ‘wake up strokes’ and when clear time of onset is unclear but LKW unlikely to be close to actual time of onset

Subacute stroke

- These patients aren't candidates for tPA or thrombectomy
- Permissive hypertension: practice varies between normotensive, 20% reduction from admission BP or allowing **SBP <220mmHg**
- Antiplatelet agents such as aspirin full dose (325mg) or clopidogrel (if aspirin intolerant) should be given within 24 hours if no contraindications
- Investigate stroke etiology
 - Echocardiogram
 - Carotid imaging
 - Telemetry monitoring
- Focus on early PT/OT/ST
- **Secondary stroke prevention**

Secondary Stroke Prevention

(for all stroke patients
whether they got
tPA/thrombectomy or not)

- All patients should be assessed for co-morbidities that place them at high risk for another stroke
 - Antiplatelet therapy: aspirin
 - Anticoagulation: for patients with atrial fibrillation, hypercoagulable states, intra-cardiac thrombus/device, mechanical valves
 - Echocardiogram
 - **30-day event monitor if no afib in hospital**
 - Blood pressure control
 - Hyperlipidemia: all stroke patients should get high dose statin therapy (atorvastatin 40mg or higher, or equivalent)
 - Glucose control: screen for diabetes
 - Smoking cessation
 - Carotid revascularization should occur within 2 weeks of index event if indicated and no contraindications
 - Carotid imaging

Malignant Hemispheric Stroke

- **MCA or combined ACA/MCA stroke in either hemisphere which involves at least 2/3 of the territory...**this is a BIG stroke
- High probability for malignant cerebral edema and brain herniation
 - High mortality
 - Consider intubation, high risk for loss of airway control
 - High risk for hemorrhagic transformation
 - Decompressive hemi-craniectomy within 48 hours in patients <60 yo to decrease mortality and improve functional outcomes, though there is great variability in this practice.

Case Example

- 7:15pm - You are notified that EMS is bringing in a suspected stroke:
 - 55 year old man
 - Hx of atrial fibrillation, hypertension, type 2 diabetes
 - Patient was with family sitting down to dinner at 7:00pm when he was witnessed to begin struggling to speak, right facial droop, unable to hold his utensils.
 - No previous neurologic deficits, was in his normal state of health immediately prior to onset of symptoms.
 - ETA 15min...

EMS will triage first to comprehensive stroke centers (thrombectomy capable,) then to primary stroke centers.

Stroke protocols will be activated, can vary from hospital to hospital but generally include notifying the ED and neurologist, as well as the interventional radiology team for possible thrombectomy.

At this point all you know is that this patient has multiple vascular risk factors with acute onset of focal neurologic deficits. (which localize pretty well)

Case Example

- Stroke team activate!
- Usually, initiated by the emergency room
- Notification goes out to neurology team, interventional radiology/neurology/neurosurgery
- Pharmacy, CT, and MRI are notified as well
- Usually, a trauma room is readied for the patient to facilitate rapid assessment

Case Example

- On arrival you quickly collect some more history from the patient's family:
 - Patient was on coumadin for the afib, but stopped the medication last week for a dental procedure
 - His hypertension and diabetes have been well managed, and he has been compliant with medications
 - No recent infections, no medication changes, he works full time as a CPA, non-smoker, social drinker.
 - Meds: lisinopril, metformin, (coumadin)
 - No recent procedures or surgeries
 - No history of clotting disorders

Case Example

→ Vitals

- HR: 85
- BP: 175/89
- Temp: 98.6
- RR: 14

Exam:

General - anxious, no acute distress otherwise

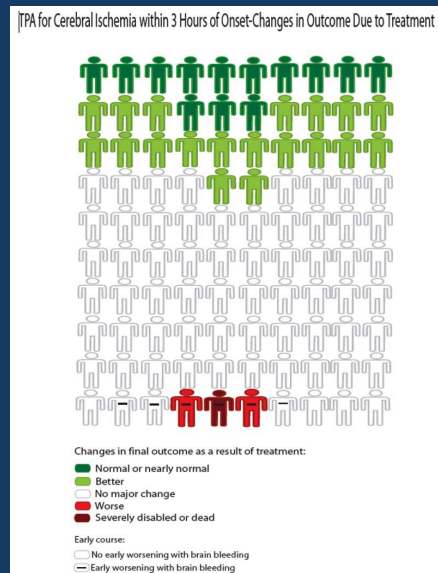
CV – irregular rate, normal S1 S2, good distal pulses,

Mental Status:

- Alert, language deficits limit orientation eval
- Speech is densely dysarthric, halting, hesitant. Naming, repetition, and comprehension are all impaired
- Leftward gaze preference
- Right homonymous hemianopia
- Right forehead-sparing facial droop
- Right pronator drift
- Right arm > leg weakness
- Right hemianaesthesia
- Right sided reflexes are slightly brisker than the left
- NIHSS - 15

Case Example

- CT Head shows no acute abnormalities
 - Clear time of onset, no contraindications
 - Push tPA!
 - Hope for green
-
- CT Angio can then be done to eval for LVO and decide on eligibility for clot retrieval



Hemorrhagic Stroke

Intracerebral Hemorrhage (ICH)

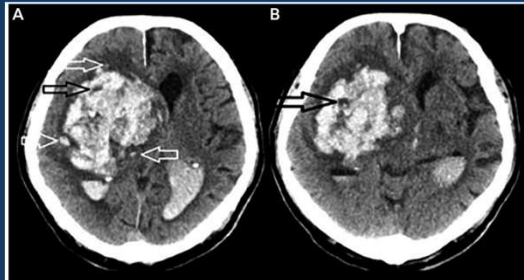


FIGURE 3-3
Imaging characteristics of intracerebral hemorrhage associated with poor prognosis. A, Axial noncontrast CT showing the island sign (ie, at least three scattered small hemorrhages separate from the main hemorrhage) (white arrows), whereas the hemorrhage has hyperdense and hypodense regions (the blend sign) (black arrow). B, Axial noncontrast CT showing the black hole sign (a hypodense region surrounded by a hyperdense region not connected with the adjacent brain tissue) (arrow).
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- Usually presents with **acute neurological deficits** and progressive clinical decline due to hematoma expansion, cerebral edema, and medical complications
- Close monitoring, prompt intervention focused on BP control, reversal of coagulopathy, and supportive care to minimize secondary injury are essential
- Differential diagnosis: ischemic stroke, TIA, SAH, subdural hematoma, meningitis, encephalitis, brain neoplasm, hypertensive emergency, migraine, acute hypoglycemia, hyper/hyponatremia

ICH pathophysiology

- 2 most common etiologies
 - Chronic arterial hypertension
 - Leads to lipohyalinosis of lenticulostriate arteries, small thalamic arteries, and pontine perforators
 - ICH occurs in **deep grey matter (basal ganglia, thalamus), pons, and cerebellum**
 - Cerebral amyloid angiopathy
 - Causes deposition of beta-amyloid in walls of small to medium sized leptomeningeal and cortical vessels
 - Elderly (>65 yo)
 - Leads to **cortical-subcortical (lobar) hemorrhages**

Clinical presentation

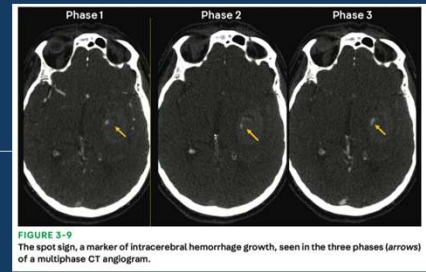
- **Focal neurological deficit depending on location**
- Depressed level of consciousness
- Headache
- Nausea/vomiting (elevated ICP, cerebellar location, hydrocephalus)
- Severe **hypertension** common
- **Seizures** associated with cortical involvement
- **Progressive clinical deterioration** in the acute phase due to ongoing hematoma expansion (in 1/3 of patients within first 3 hours) – things get bad FAST

Risk factors/triggers

- Most common trigger: noncompliance with antihypertensive medication and chronic hypertension
- Physical activities that raise BP
- Supratherapeutic INR
- Sympathomimetic drug use (cocaine, amphetamines)
- Trauma
- Prior stroke
- Male gender
- Age >55
- Heavy alcohol consumption
- Liver dysfunction
- Peripartum period

Work up

- **Neurological emergency** – any level of suspicion should lead to expedited assessment
- Obtain brief medical history from patient or bystander on time of onset, vascular risk factors, history of htn, medications, illicit drug use, recent head trauma, history of prior stroke, and baseline cognitive status
- Labs: urine drug screen, chemistry, cbc, pt/ptt, anti-Xa level
- Imaging: **non contrast head CT** (sensitivity/specificity near 100% in 1st week)
 - Adding contrast (CTA) may permit visualization of ‘**spot sign**’ representing active extravasation into hematoma – associated with expansion and increased mortality, and may identify underlying vascular anomalies



Treatment/management principles

- ICU admission: risk of early deterioration and cardiorespiratory compromise
- Blood pressure management
 - Current AHA guidelines recommend targeting SBP <140mmHg
 - May require continuous infusion of IV antihypertensives (nicardipine, clevidipine)
- Coagulopathy management
 - Discontinue antiplatelet or anticoagulant medications
 - **Warfarin reversal:** vitamin K + prothrombin complex concentrate (kcentra) preferred over fresh frozen plasma (FFP)
 - **Idarucizumab for dabigatran reversal**
 - **Routine platelet transfusion should be avoided in spontaneous ICH**

Treatment/management principles

→ Treatment of elevated ICP and hydrocephalus

- Hydrocephalus complicates up to 50% of cases
- More common in younger patients, with lower GCS, and deep hemorrhages
- Consider placement of an ICP monitor if GCS <8, evidence of herniation, extensive interventricular hemorrhage or hydrocephalus

→ Seizures

- No prophylaxis for spontaneous or non traumatic ICH
- Treat seizures if they occur
- Low threshold for cEEG monitoring in cases of depressed mental status out of proportion to ICH – rule out non convulsive status epilepticus

Treatment/management principles

→ Neurosurgical intervention

→ Current evidence does not show benefit of prophylactic early surgery in the setting of supratentorial ICH

*in contrast to malignant mca infarct

→ Patients with **cerebellar hemorrhage >3cm** in diameter or with **symptomatic mass effect** should have **emergent surgical decompression to prevent herniation!**

→ Fever Control

→ Fever is common in ICH and associated with worse outcomes

→ **Treat fevers >38.5C** – acetaminophen, chilled saline infusion, cooling blankets

Prognosis

- Mortality from ICH remains high (40% at 30 days)
 - depends on patient, hematoma location, volume, active extravasation.
- Only 12-39% of patients achieve functional independence post discharge – **high morbidity!**
- **ICH score: ranges from 0-6**
 - Higher score = higher 30-day mortality
 - Remember that initial aggressive care is indicated for all patients regardless of score – otherwise it's a self-fulfilling prophecy!
 - Current guidelines discourage withdrawal of life sustaining treatments in the first 24 hours

Component	ICH Score Points
GCS score	
3-4	2
5-12	1
13-15	0
ICH volume (cm ³)	
≥ 30	1
< 30	0
IVH	
Yes	1
No	0
Infratentorial origin of ICH	
Yes	1
No	0
Age (year)	
≥ 80	1
< 80	0
Total ICH Score	0-6

The GCS score refers to the GCS score at initial presentation (or after resuscitation); ICH volume, volume on initial CT calculated using the ABC/2 method; IVH, presence of any IVH on the initial CT.

GCS, Glasgow coma scale; ICH, intracerebral hemorrhage; CT, computed tomography; IVH, intraventricular hemorrhage.

Adapted from Hemphill JC 3rd, Bonovich DC, Besmertis L, Manley GT, Johnston SC. The ICH score: a simple, reliable grading scale for intracerebral hemorrhage. *Stroke* 2001;32:891-897.

Subarachnoid Hemorrhage

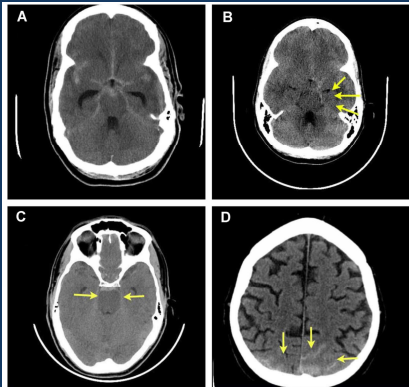


Fig. 1. Patterns of hemorrhage. (A) Obvious aneurysmal SAH. (B) More subtle SAH. (C) Perimesencephalic hemorrhage. (D) Convexal SAH. (Courtesy of J. Edlow, MD.)
Dukech NM and Edlow JA. PMID: 33218684

- Neurological emergency associated with significant morbidity and mortality
- Outcomes greatly impacted by early recognition and aggressive treatment
- 2 main etiologies
 - Traumatic
 - Non traumatic
 - **Aneurysmal (80%) - aSAH**
 - Non aneurysmal

Epidemiology / Risk Factors

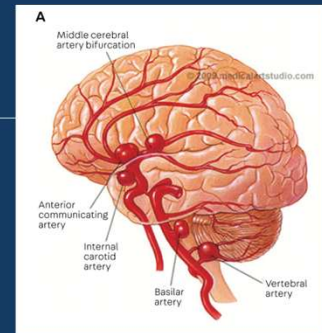
- 3rd most common cerebrovascular disorder after ischemic stroke and ICH
- Based on autopsy studies – prevalence of unruptured intracranial aneurysms in adults is 1-5%
- Worldwide incidence of aSAH 2-16/100,000 adults. In the US 9-14/100,000 adults
- Higher incidence in women (2:1), African-Americans, Hispanics, and age >55 yo
- Risk factors:
 - Hypertension, smoking, alcoholism.
 - Family history of SAH or another aneurysm.
 - **Autosomal Dominant Polycystic kidney disease**, Ehlers-Danlos syndrome, fibromuscular dysplasia, Marfan's syndrome
 - Sympathomimetic use (e.g., cocaine, amphetamine).

Pathophysiology

- Occurs due to **extravasation of blood into subarachnoid space** under high hemodynamic pressure
- 80% of spontaneous, non traumatic SAH is due to **aneurysm rupture**
 - Aneurysms at vessel bifurcations due to defects in tunica media and elastica - pathologic dilations at major branching brain arteries
 - Other theories include inflammatory process affecting cerebral vasculature or disruption of internal elastic membrane from hemodynamic forces
- Non-aneurysmal causes:
 - Rupture of dissections
 - Arterial blisters
 - Trauma
 - Uncommon causes: vascular malformations, coagulopathy, vasculitis, moyamoya, amyloid angiopathy, sympathomimetic exposure

Clinical presentation

- **Thunderclap headache** – ‘worst headache of my life’
 - Some patients will also have ‘sentinel headache’ = warning headache preceding aSAH by days or weeks
- **Meningeal signs:** photophobia, nuchal rigidity, neck pain
- **Elevated Intracranial Pressure:** nausea, vomiting, stupor, coma
- **Seizures:** blood is a cortical irritant
- **CN palsies/focal neurological deficits:** due to direct compression from aneurysm or blood
 - Posterior communicating artery: CN III palsy with pupillary involvement
 - MCA: contralateral face and arm paresis, aphasia, visual neglect
 - ACA: bilateral leg paresis and Babinski sign
 - Basilar tip: vertical gaze paresis, bitemporal hemianopsia, bilateral motor paresis, coma



Diagnosis / Work up

- Non contrast head CT: initial test of choice
 - 99% sensitive within 6 hours of symptom onset (93% at 24 hours, 50% at 1 week)
- CT angiography: useful to evaluate aneurysm location and morphology
 - CTA unreliable for small <3mm aneurysms
- Digital subtraction angiography (DSA): GOLD STANDARD imaging for cerebral vasculature – however this is an interventional neurology procedure
 - Used for surgical planning and intervention
- Lumbar puncture: almost 100% sensitive when done >12 hours after symptom onset
 - For patients with negative non contrast CT head but high suspicion of SAH
 - **Observe for xanthochromia** – yellow appearance of spun down CSF
- MRI/MRA: not used in acute period due to time requirements

Grading Scales

	Hunt and Hess	World Federation of Neurosurgical Societies (WFNS)	Modified Fisher Score based on CT scan
1-3 = Good Grade 4-5 = Poor Grade	Grade 1 Asymptomatic or mild headache Minimal (if any) nuchal rigidity Risk of mortality 30%**	GCS 15	Thin* SAH, no intraventricular hemorrhage Risk of symptomatic vasospasm: ~10%
	Grade 2 Moderate to severe headache Nuchal rigidity No neurologic deficit (other than CN palsy) Risk of mortality 40%**	GCS 13-14 & no motor deficit	Thin* SAH with intraventricular hemorrhage Risk of symptomatic vasospasm: ~20%
	Grade 3 Lethargy or confusion Mild focal deficit Risk of mortality 50%**	GCS 13-14 & motor deficit	Thick* SAH, no intraventricular hemorrhage Risk of symptomatic vasospasm: ~30%
	Grade 4 Stupor Moderate to severe hemiparesis Risk of mortality 80%**	GCS 7-12	Thick* SAH with intraventricular hemorrhage Risk of symptomatic vasospasm: ~40%
	Grade 5 Deep coma Signs of severe neurological impairment (e.g., posturing) Risk of mortality 90%**	GCS 3-6	

*Thick is defined as filling one or more cisterns or fissures, out of a total of 10 cisterns/fissures: interhemispheric fissure, the quadrigeminal cistern, both suprasellar cisterns, both ambient cisterns, both basal sylvian fissures, and both lateral sylvian fissures.
 **Mortality rates are somewhat outdated, so the risk of mortality with modern management may be lower. Additionally, it should be noted that SAH grades can change over time and following therapy.

The Internet Book of Critical Care, emcrit.org/ibcc/sah

- Clinical and radiographic scales for likelihood of developing complications and outcome prognosis
 - Clinical: Hunt and Hess, World Federation of Neurological Surgeons
 - Radiographic: modified Fisher scale

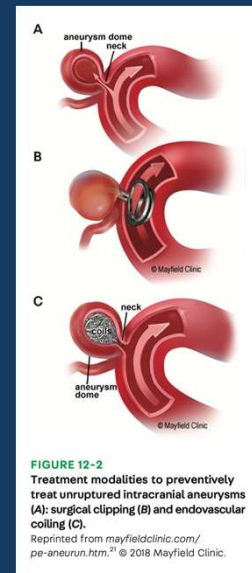
Treatment / management principles

→ **Secure the aneurysm!**

- Clipping: Craniotomy with surgical ligation
- Coiling: endovascular embolization with coils

→ BP management

- Before the aneurysm is secured control hypertension and keep SBP <140 mmHg
- Avoid hypotension that may result in secondary brain injury from hypoperfusion



Treatment/ management principles

- Hydrocephalus (15-20% of patients with SAH)
 - Patients will present with decreased level of consciousness, usually have interventricular hemorrhage
 - Requires placement of an External Ventricular Drain (EVD)
- Re-bleeding
 - Occurs in 4% of patients in first 24 hours if aneurysm is unprotected
 - Highest risk in 1st day, but up to 2 weeks out
 - Prevent by securing the aneurysm within first 24 hours
- **Vasospasm** – delayed cerebral ischemia (DCI)
 - Can occur up to 21 days after SAH
 - Patients will have sudden neurological deficits on exam
 - Nimodipine (calcium channel blocker) is only approved drug therapy to reduce vasospasm
 - May require intra-arterial drug therapy and/or angioplasty

Treatment / management principles

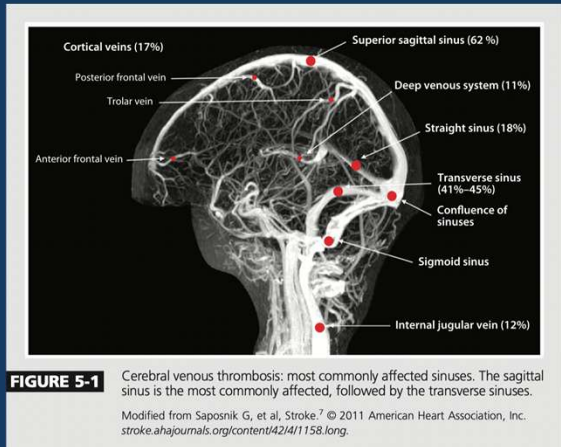
- Sodium and volume management
 - Hyponatremia and hypovolemia are common after SAH and increase risk of complications
 - Goal is EUNATREMIA and EUVOLEMIA
- Seizures
 - 5-20% of patients after SAH
 - May be the presenting symptom in 8% of patients
 - Prophylactic anti-epileptic drugs (AEDs) are controversial
 - Treat seizures if they occur and then continue AEDs for the duration of hospitalization
 - Low threshold for cEEG to rule out non convulsive status in patients with depressed mental status

Outcomes

- ~20% of SAH patients treated at experience high volume centers do not survive to discharge
- Survivors are disabled by cognitive impairment, long term problems in memory, concentration, executive function, depression, anxiety
- Cognitive and physical rehabilitation essential for maximizing recovery

Venous Thrombosis

Cerebral Venous Thrombosis



- Thrombosis of the dural sinus and/or cerebral veins (CVT)
- most commonly affected venous sinus is the superior sagittal (62%) followed by transverse sinus (41% to 45%)
- Causes dysfunction by 2 mechanisms:
 - (1) **Localized edema** of the brain and venous infarction.
 - (2) Thrombosis of the major sinuses leads to **intracranial hypertension** as a result of increased venous pressure and impaired absorption of CSF - **hydrocephalus**.

CVT Epidemiology & Risk Factors

- Rare type of stroke
 - 0.5% of patients with strokes
 - Primarily younger individuals age <50 yo, median age 37 yo
- 75% of CVT cases are in women
- Incidence in children is 0.67 per 100,000 per year
 - 43% of those in neonates
- Risk Factors: Female Sex (3:1)
 - 65% of identifiable causes of CVT in women related to a risk factor unique to women:
 - oral contraceptives, pregnancy, post partum period, and hormone therapy.

TABLE 5-3 Common Underlying Causes of Cerebral Venous Thrombosis or Dural Sinus Thrombosis Designated by Transient Versus Chronic/High-Risk Triggers

Condition	Prevalence (%) ^a	Trigger Type
Prothrombotic conditions	34.1	Chronic ^b
Pregnancy and puerperium	21.0	Transient
Oral contraceptives/hormone replacement	54.3	Transient
Drugs (eg, danazol, lithium, vitamin A, IV immunoglobulin, ecstasy)	7.5	Transient
Cancer related	7.4	Transient or chronic
Local compression		
Hypercoagulability		
Antineoplastic drugs (eg, tamoxifen, L-asparaginase)		
Infection	12.3	Transient
Parameningeal (eg, ear, sinus, mouth, face, and neck)		
Sepsis		
Mechanical precipitants	4.5	Transient
Complication of epidural blood patch		
Spontaneous intracranial hypotension		
Lumbar puncture		
Dehydration	No prevalence data but is a recognized trigger	Transient
Other hematologic disorders	12.0	Chronic
Paroxysmal nocturnal hemoglobinuria		
Iron deficiency anemia		
Thrombocytopenia		
Nephrotic syndrome		
Polycythemia		
Systemic diseases	7.2	Chronic
Systemic lupus erythematosus	1.0	
Behçet disease	1.0	
Inflammatory bowel disease	1.6	
Thyroid disease	1.7	
Sarcoidosis	0.2	
Other	1.7	
None identified	12.5	Unknown

^a Prevalence rates among patients with CVT from the International Study of Cerebral Venous Thrombosis.¹

^b Excludes the acquired and potentially transient deficiencies associated with the acute phase of systemic conditions (eg, nephrotic syndrome).

Clinical presentation

- (1) isolated **intracranial hypertension**:
 - Headache
 - Diplopia
 - visual impairment, severe papilledema
 - decreased level of consciousness
 - sixth nerve palsy
- (2) **focal neurologic deficit** (eg, motor weakness, sensory deficit, aphasia)
- (3) **encephalopathy**
- (4) **seizures** (with or without a focal neurologic deficit)

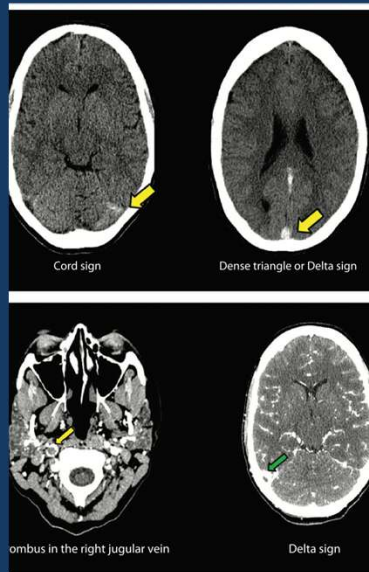
TABLE 5-1 Clinical Manifestations of Cerebral Venous Thrombosis

Clinical Presentation	Frequency ²
Headaches	90%
Seizures	40%
Focal deficit (eg, hemiparesis, aphasia)	20%
Decreased level of consciousness	14%
Isolated headaches ⁵	15%
Visual loss	13%

Diagnosis

- High level of clinical suspicion needed
- Labs: d-dimer
 - Elevated in acute phase or with extensive clot
 - In subacute/chronic phase low d-dimer doesn't rule out CVT
- Imaging:
 - CT brain
 - CT venogram
 - MRI brain
 - Magnetic resonance venogram (MRV)
 - Conventional angiogram

CT head



CT venogram



MRV

Acute Management

→ Anticoagulation:

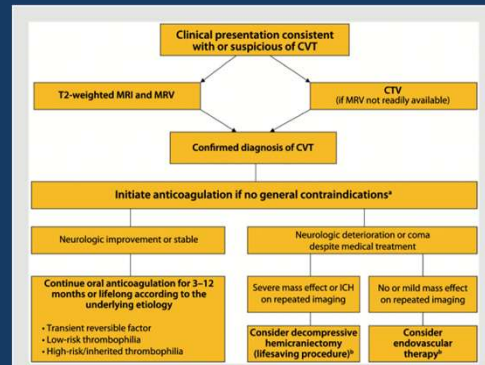
- Limited evidence from large RCTs
- Used to prevent thrombus growth, to facilitate recanalization, and to prevent other thrombotic events such as deep venous thrombosis and pulmonary embolism

→ Endovascular Therapy: thrombectomy or thrombolysis

- For patients who develop progressive neurologic deterioration despite intensive medical treatment

→ Decompressive Hemicraniectomy

- Should be reserved for young patients who develop a malignant hemispheric stroke due to CVT with progressive neurologic deterioration despite intensive medical treatment.



Management of complications

Seizures	Hydrocephalus	Intra-cranial Hypertension	Visual Loss
• May occur in 1/3 of adults and nearly 1/2 of children with CVT • Factors associated with development of seizures: venous infarction, hemorrhagic conversion, and intracranial hypertension • No evidence for prophylactic use of antiepileptic drugs • Guidelines recommend early initiation of antiepileptic drugs in patients 1 or more seizures	• Thrombosis of major sinuses (eg, superior sagittal, transverse sinuses) may affect CSF drainage, resulting in communicating hydrocephalus • Treatment includes measures to decrease the production of CSF, recanalization of the thrombosed sinus, and management of intracranial hypertension. • Neurosurgical consultation for evacuation of CSF either with ventriculostomy or ventriculoperitoneal shunt may be necessary	• Seen in 15%-40% of patients with CVT • Caused by venous outflow obstruction and CSF malabsorption. • Treatment: No RCTs have tested these interventions • Diuretics (ie, acetazolamide) • Serial lumbar punctures • In refractory cases, lumboperitoneal shunt may be required	• Due to pressure on the optic nerves • Reported in up to 13% of patients • Treatment: • Ophthalmic consultation • Monitor for papilledema • Visual field testing • For patients at high risk of blindness, optic nerve fenestration may be considered

Prognosis

- Outcomes after CVT are generally favorable
- Women do better than men if they had a sex-specific risk factor
 - In one study 81% of women had complete recovery versus 71% of men
 - Women without sex-specific risk factors tended to have a worse outcome than women with these risk factors (OR for poor outcome 3.7; 95% CI, 1.9 to 7.4)
- Risk of any thrombotic event after a CVT is about 6.5% per year (DVT, PE, Stroke, etc)
- Chronic anticoagulation: Depends on risk factors
 - Provoked CVT: 3-6 months of anticoagulation
 - Unprovoked CVT: 6-12 months of anticoagulation warfarin
 - Recurrent CVT, venous thromboembolism after CVT, or first CVT (provoked or unprovoked) with severe thrombophilia: indefinite anticoagulation may be considered